MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course(VAC) - MG3VACTXLA00 - TEXTILE PROCESSES
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A – MCQ
Answer all questions
Each question carries 1 mark

1. Which type of singeing is most commonly used in textile industry?
a) Gas singeing
b) Plate singeing
c) Roller singeing
d) Chemical singeing
Answer: a) Gas singeing

2. The most widely used bleaching agent for cotton is:
a) Sodium hypochlorite
b) Hydrogen peroxide
c) Sulphur dioxide
d) Potassium permanganate
Answer: b) Hydrogen peroxide

3. What is the primary purpose of applying a stain-resistant finish to textile fabrics?
a) To prevent liquids and dirt from penetrating the fabric
b) To improve fabric colorfastness
c) To enhance thermal insulation
d) To increase fabric weight
Answer: a) To prevent liquids and dirt from penetrating the fabric

4. Name what types of fibers dope dyeing is used for?
a) Natural fibers
b) Synthetic fibers
c) Blended fibers
d) All fibers
Answer: b) Synthetic fibers

5. Dip dyeing is best suited for.
a) Knits only
b) Garment dyeing and artistic designs
c) Industrial polyester dyeing
d) Roller printing
Answer: b) Garment dyeing and artistic designs

6. Tumble dyeing is carried out in.
a) Roller printing machine
b) Rotating drum
c) Kier boiler
d) Padding mangle
Answer: b) Rotating drum

7. At which stage Top dyeing is being carried out?
a) Fiber stage
b) Sliver/top stage
c) Fabric stage
d) Garment stage
Answer: b) Sliver/top stage

8. Cross dyeing produces.
a) Solid uniform color
b) Multicolor effects on blended fabrics
c) Only black shades
d) Colorless fabrics
Answer: b) Multicolor effects on blended fabrics

9. The tool used to push paste through the screen is called.
a) Hammer
b) Squeegee
c) Roller
d) Brush
Answer: b) Squeegee

10. Rotary printing uses rotating _____ screens.
a) Flat
b) Cylindrical
c) Round
d) Cube
Answer: b) Cylindrical

Part B – Fill in the blanks
Answer all questions
Each question carries 1 mark

11. _____ mercerization is carried out under tension to avoid shrinkage.
Answer: Tension mercerization

12. The removal of starch or other size material from fabric is called _____.
Answer: Desizing

13. Flame retardant finishes reduce the fabric's tendency to _____ when exposed to heat.
Answer: burn

14. Beam dyeing is limited to _____ weight fabrics.
Answer: light

Part B – Fill in the blanks (continued)

Each question carries 1 mark

15. A disadvantage of stock dyeing is its high _____.
Answer: price

16. Jet dyeing requires a _____ liquor ratio, saving water and energy.
Answer: low

17. Skein dyeing produces yarn that is _____ in texture and suitable for hand-knitting.
Answer: loose

18. Pad dyeing requires further fixation steps like steaming, thermofixation, or _____ treatment.
Answer: chemical (or) padding

19. The two components used in naphthol dyeing are naphthol and _____.
Answer: diazo salt

20. In block printing, design is transferred by repeated _____ of the block.
Answer: pressing (or) stamping

Part-C One sentence answer
Each question carries 1 mark

21. Define preparatory processes in textile finishing.

Answer: Preparatory processes are the treatments given to textiles before dyeing or finishing to remove impurities and improve absorbency.

22. Mention two uses of flocked fabrics.

Answer: Flocked fabrics are used in upholstery and garment decoration.

23. What is the process called that controls shrinkage in cotton by mechanical compression?

Answer: The process is called sanforization.

24. Which fibers are commonly dyed using acid dyes?

Answer: Wool and silk fibers.

25. Mention two fibers suitable for basic dyes.

Answer: Acrylic and nylon.

26. Mention two advantages of reactive dyes.

Answer: Reactive dyes have good wash fastness and bright shades.

27. What is yarn stage dyeing?

Answer: Yarn stage dyeing is the process of dyeing yarns before they are woven or knitted into fabric.

28. What are the common sources of natural dyes?

Answer: Natural dyes are obtained from plants, insects, and minerals.

29. What is jig dyeing?

Answer: Jig dyeing is a process where fabric passes back and forth through a dye bath under tension.

30. Mention two disadvantages of discharge printing.

Answer: It uses harmful chemicals and is suitable only for specific dyes.

Part D- Short answer
Answer any 5 questions
each question carries 2 marks

31. Name two types of fabrics that commonly undergo shearing and explain why.

Answer: Woolen fabrics and velvet undergo shearing to remove surface fuzz and create a smooth, even finish.

32. How does a water repellent finish differ from a waterproof finish?

Answer: A water repellent finish allows water to penetrate the fabric but resists wetting, whereas a waterproof finish completely prevents water from passing through the fabric.

33. Why are vat dyes considered superior for denim dyeing?

Answer: Vat dyes are considered superior for denim dyeing due to their excellent colorfastness properties, particularly to washing and light.

34. State any two limitations of winch dyeing.

Answer: Two limitations of winch dyeing are: 1) It can cause uneven dyeing if not properly controlled, and 2) It may lead to fabric creasing.

35. Mention two applications of direct dyes.

Answer: Two applications of direct dyes are: 1) Dyeing cotton and other cellulosic fibers, and 2) Dyeing wool and silk.

36. Mention two advantages of roller printing.

Answer: Two advantages of roller printing are: 1) High production rate, and 2) Ability to produce intricate designs.

Long-answer questions (Part E)
Answer any two question
Each question carries 5 marks

37. Discuss at least three advantages of napping and explain how they improve the performance or appeal of fabric.

Answer: Napping improves fabric softness, warmth, and aesthetic appeal. It also enhances the fabric's ability to retain warmth and provides a plush texture.

38. Discuss the advantages, limitations, and applications of disperse dyes.

Answer: Disperse dyes are used for dyeing synthetic fibers like polyester. Advantages include good colorfastness and a wide range of colors. Limitations include difficulty in achieving deep shades and potential environmental concerns. Applications include dyeing polyester fabrics.

39. What are the advantages and disadvantages of package dyeing?

Answer: Advantages of package dyeing include efficient use of dye and uniform coloration. Disadvantages include potential for uneven dyeing if packages are not properly wound and the need for careful control of dyeing conditions.

40. Explain the process of duplex printing.

Answer: Duplex printing involves printing designs on both sides of a fabric simultaneously using two rollers, one for each side. This process allows for intricate designs on both sides of the fabric with good registration.